

Lesson 1 · Practice sheet

Name: _____

Date: _____

Show all working. Write $f'(x)$ every time you use calculus. The letter at the right shows the level: [A] Achieved, [M] Merit, [E] Excellence.

1. Differentiate $f(x) = 6x^4 - 3x^3 + 2x - 11$. [A]

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2. Find the gradient of $y = x^3 - 4x^2 + 5$ at the point where $x = -1$. [A]

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3. Find the coordinates of the point on $f(x) = 3x^2 - 12x + 2$ where the gradient is 6. [A]

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4. Find the equation of the tangent to $y = x^3 - 3x^2 + 2$ at $x = 3$. [A]

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Question 5 [A]

- (a) A function f is given by: $f(x) = 4x^3 - 2x^2 - 7x + 4$

Use calculus to find the gradient of the graph of the function at the point where $x = 3$

Question 6 [A]

- (b) For the cubic function $f(x) = \frac{1}{2}x^3 + \frac{1}{2}x$ find the equation of the tangent to the curve at $x = 2$

Question 7 [A] · Merit if all steps are linked

- (b) The function f is given by: $f(x) = 5 + 3x + cx^2 - 2x^3$

At the point on the graph of the function where $x = 2$, the gradient is -5 .

Find the value of c .

Question 8 [A]

- (a) A function f is given by $f(x) = 2x^4 + 4x^3 - 20x^2 - 5$.

Use calculus to find the gradient of the graph of the function at the point where $x = 3$.

Question 9 [M]

- (d) The tangent to the curve $f(x) = px - qx^2$ at the point $(2, -10)$ has a gradient of -6 .

Find the values of the constants p and q .

Question 10 [A]

- (a) Find the equation of the tangent of the curve $y = 2x(x - 3)$, where $x = 1$.
